Composing the Stratified and Lateral Propagators

Reciprocity Gates, the Source-Convention Reconciliation, and the Case for a Depth-Stratified Crust Reference

26 July 2026

Abstract

The thesis formulates heterogeneous ocean crust as a *two-potential* (distorted-wave) problem: a laterally invariant, depth-dependent reference solved exactly, plus a lateral contrast summed as a multiple-scattering series *carried by the reference Green's tensor.* Realising this numerically requires composing a stratified vertical propagator $\mathscr{P}^z$ with a disorder-resolved lateral propagator $\mathscr{P}^x$ in a single resolvent. This note records what had to be established before that composition was legitimate. Four results: (i) the two-stage solve is *algebraically identical* to one global Foldy–Lax solve with $\mathscr{P}^x + \mathscr{P}^z$; (ii) the crust gradient must live in the reference, not in the scattering series — a $\pm 0.5\%$ velocity gradient changes the inter-scatterer coupling by 10%; (iii) the two available propagators use *different source representations*, a state jump versus a body force, related by one factor of the system matrix; and (iv) with that conversion and an adjoint source operator, both satisfy a common reciprocity invariant to machine precision. Six errata in the thesis were found and corrected along the way.

Contents

1 The object being built

The thesis splits the medium into a laterally invariant, depth-dependent reference — the depth average, which generally increases with depth — plus a lateral contrast superimposed as cylinders. The first potential is solved exactly by Kennett-like matrix recursions; the second is summed as a multiple-scattering series. The master result of Chapter 5 is

$$\mathscr{G} = {}_r\mathscr{G} + \mathscr{P}_s^{z,R}\,\Delta\mathscr{C}_{\text{eff}}\Big(\mathbf{I} - \big[\mathscr{P}^z + \mathscr{P}^x\big]\Delta\mathscr{C}_{\text{eff}}\Big)^{-1}{}_r\mathscr{G}_r^z, \tag{1}$$

with $\mathscr{Q}^\partial \equiv (\mathbf{I} - \mathscr{S}_{\text{int}}\mathscr{E})^{-1}\mathscr{S}_{\text{int}}$ the extension of the plane-wave propagator to a stratified medium, and $\mathscr{P}^z \equiv -i\,\Theta^z_{\text{in}}\mathscr{Q}^\partial\Theta^z_{\text{ou}}$. Because $\mathscr{Q}^\partial$ depends only on the reference, it is built once per frequency and every Krylov iterate is then a cheap sweep. That economy is the point of the splitting.

1.1 The two-stage solve is exact

The thesis first solves each row internally, $\Delta\mathscr{T} = \Delta\mathscr{C}_{\text{eff}}(\mathbf{I} - \mathscr{P}^x\Delta\mathscr{C}_{\text{eff}})^{-1}$, and then couples rows through $\mathscr{P}^z$. The collapse to (1) is a resolvent identity: with $A(\mathbf{I} - PA)^{-1} = (A^{-1} - P)^{-1}$ and $\Delta\mathscr{T}^{-1} = \Delta\mathscr{C}_{\text{eff}}^{-1} - \mathscr{P}^x$,

$$\Delta\mathscr{T}\big(\mathbf{I} - \mathscr{P}^z\Delta\mathscr{T}\big)^{-1} = \big(\Delta\mathscr{C}_{\text{eff}}^{-1} - \mathscr{P}^x - \mathscr{P}^z\big)^{-1} = \Delta\mathscr{C}_{\text{eff}}\Big(\mathbf{I} - \big[\mathscr{P}^x + \mathscr{P}^z\big]\Delta\mathscr{C}_{\text{eff}}\Big)^{-1}. \tag{2}$$

Solving rows first and then stacking them is algebraically identical to one global Foldy–Lax solve with $P = \mathscr{P}^x + \mathscr{P}^z$. The row/stack division is a computational strategy, not different physics. Verified numerically at $n = 1, 3, 9, 27$ to 3×10^{-16}, and independently against the global Neumann series $\sum_k \Delta\mathscr{C}_{\text{eff}}(P\Delta\mathscr{C}_{\text{eff}})^k$ to 4.7×10^{-16}.

This is what licenses building $\mathscr{P}^x$ and $\mathscr{P}^z$ separately and adding them.

2 The crust gradient belongs in the reference

An earlier design made the crust reference a single homogeneous block, with the internal crust interfaces transparent, purely to reduce cost. That choice moves the depth gradient out of the reference and into the scattering series. Its size is measurable.

Holding the geometry fixed — same ocean, seabed, half-space and interface depths — and varying only whether the crust reference carries its gradient, the relative change in the inter-scatterer coupling $G_9(i\leftarrow j)$ is:

crust α gradient	complex difference	magnitude only
4.00–4.00 (uniform)	0.0%	0.0%
3.98–4.02 (±0.5%)	**10.5%**	8.4%
3.95–4.05	26.5%	23.2%
3.90–4.10	55.6%	50.3%
3.00–5.00 (realistic, 67%)	∼130%	∼35%

Controls: the null test (identical models) returns exactly zero, and the difference grows smoothly from zero with gradient strength, so this is physics rather than a coding artefact. Across $f = 5, 10, 20$ Hz and horizontal slowness $p = 0$–0.2 s/km the realistic case runs 70–340% uniformly: the operator is *different*, not perturbed.

A ±0.5% velocity gradient across the crust changes the coupling by 10% because the effect is a travel-time accumulation along the path — the "habitual mistiming" the thesis quotes Coates & Charrette on, and precisely what the two-potential split exists to remove. **The uniform-host crust is not a small correction and should not be used as a cost saving.**

3 The source-convention obstruction

Two 9-component propagators are available. Both claim the state $(u_z, u_x, u_y,\ \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy},\ 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx})$, but they are built from opposite ends, and the difference is *not* a normalisation.

3.1 A calibrated invariant

Reciprocity is the right discriminator: it holds in any elastic medium, so no uniform limit or $k \leftrightarrow r$ transform is needed, and it is sensitive to exactly the source-side weighting at issue.

For the validated closed-form propagator $P = [[G, C], [H, S]]$ the elementwise ratio P_{ab}/P_{ba} is an *exact* outer product w_a/w_b with $w = (1, 1, 1, 1, 1, 1, 2, 2, 2)$ — the Voigt engineering factor, strain rows carrying 2ε on the shears while stress columns carry plain σ. Hence the invariant

$$\mathbf{W}P \text{ is symmetric,} \qquad \mathbf{W} = \mathrm{diag}\big(1, 1, 1,\ 1, 1, 1,\ \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}\big), \tag{3}$$

satisfied to 4.2×10^{-18}. This was *measured, not assumed*: the first version of the check asserted plain symmetry and failed at 10–16%. A gate whose calibration leg is not itself checked is worthless.

3.2 The layered propagator obeys a different law

The layered 9×9 fails (3) at ∼110%, and the failure survives every diagonal rescaling. Dropping to the 6×6 core, its exact law is

$$M(i \leftarrow j)(+k) = \frac{i}{\omega}\, \mathbf{SD}\, \mathbf{J}_6 \big[M(j \leftarrow i)(-k)\big]^T \mathbf{J}_6\, \mathbf{SD}, \qquad \mathbf{SD} = \mathrm{diag}\big(1, -1, -1,\ -i\omega,\ +i\omega,\ +i\omega\big), \tag{4}$$

verified to 6×10^{-16}. It factors into $\mathrm{diag}(1, 1, 1, -i\omega, -i\omega, -i\omega)$ — the $\sigma/(-i\omega)$ mixed basis, applied uniformly to P–SV and SH — times $\mathrm{diag}(1, -1, -1, 1, -1, -1)$, the parity of the x, y components under $k \to -k$. The symplectic signature is required: the joint rank-one fit residual is 4.8×10^{-15} with $\mathbf{J}_6$ and 0.97 with the identity.

A trap. Recovering the SH entries *requires* $k_y \neq 0$. At $k_y = 0$ the SH channel decouples, components 2 and 5 vanish identically, and a least-squares fit returns an unconstrained value for them — in our case a plausible-looking $\sqrt{\omega}$ that is simply wrong.

3.3 Translation invariance clears the solver

Before treating this as a convention issue it had to be ruled out as a defect, since a genuinely non-reciprocal Green's function would invalidate everything built on it. In a homogeneous medium $G(i \leftarrow j)$ must depend only on $z_i - z_j$, and *a constant basis weight cannot break that* — the same weights appear at every interface and cancel. The test therefore probes the Green's function alone. Two independent knobs, both monotone to zero with no plateau: attenuation $Q = 3 \to 0.5$ gives $6.0 \times 10^{-2} \to 5.6 \times 10^{-4}$; moving the test pairs deeper at fixed Q gives $2.5 \times 10^{-1} \to 1.7 \times 10^{-5}$. The residual is free-surface and seabed leakage behaving exactly as leakage must. **The layered solver is sound.**

4 The conversion: state jump versus body force

The obstruction is structural. The layered Green's function injects a unit *jump in the displacement–traction state*; the closed-form propagator takes a *(force, stress-glut)* source. These are different objects, and no rescaling converts one into the other.

Integrating $\partial_z b = \mathbf{A}b + F_1\delta(z - z_S) + F_2\delta'(z - z_S)$ across z_S: a δ' term produces a state jump $[b] = F_2$, a δ term produces none. With the Box 5.3 projection $\Sigma = -i\,\mathbf{D}_z(-k)^T\mathbf{J}_6\,(F_1 + \mathbf{A}F_2)$ this gives

$$\boxed{\text{state jump } S \;\equiv\; \text{body force } F = \mathbf{A}\,S} \tag{5}$$

where $\mathbf{A}$ is the Fourier-domain system matrix. Since $\|\mathrm{offdiag}\,\mathbf{A}\| = 5.6 \times 10^6$, the conversion genuinely *mixes* components — which is why no diagonal weight could ever have reconciled the two conventions. Derivation and the earlier numerical experiment agree.

Validated headless in Mathematica against the thesis's own identities: the quasi-Hamiltonian relation $\mathbf{A}^T(-k)\mathbf{J}_6 + \mathbf{J}_6\mathbf{A}(k) = 0$ to 0; eigenvalues $= i\,\mathrm{diag}(\pm k_z)$ to 0; $\mathbf{A}\mathbf{D}_z = \mathbf{D}_z\Lambda$ against the energy-normalised eigenbasis to 7×10^{-12}; and Box 5.3's own worked point-explosion example reproduced exactly, both the intermediate $F_1 + \mathbf{A}F_2$ and the closed-form $\Sigma \propto (1, 0, 0, 1, 0, 0)$ — P only, equal up and down, as an explosion must radiate.

5 The reconciliation

Two corrections suffice, and both are forced rather than fitted.

(1) Source normalisation. Requiring the corrected 6×6 to obey clean symplectic reciprocity $N_1 = \mathbf{J}_6 N_2^T \mathbf{J}_6$, and matching against (4) using $\mathbf{J}_6\,\mathrm{diag}(d_A, d_B)\mathbf{J}_6 = -\,\mathrm{diag}(d_B, d_A)$, determines the source-side factor uniquely up to the receiver-side freedom. Taking the receiver side as the identity,

$$N = M\,\mathbf{Q}, \qquad \mathbf{Q} = \mathrm{diag}\Big(1, -1, -1,\ \tfrac{i}{\omega}, -\tfrac{i}{\omega}, -\tfrac{i}{\omega}\Big), \tag{6}$$

after which the 6×6 obeys clean symplectic reciprocity to 6.5×10^{-16}.

(2) Adjoint source operator. The existing source operator is plain Hooke's law and wavenumber-independent, while the receiver operator $A(k)$ is not. Requiring $A_1 N_1 B_1 = \mathbf{W}^{-1}(A_2 N_2 B_2)^T \mathbf{W}$ given $N_1 = \mathbf{J}_6 N_2^T \mathbf{J}_6$ forces

$$B(k) = -\,\mathbf{J}_6\, A_{\mathrm{src}}(-k)^T\, \mathbf{W}, \tag{7}$$

built with the *source* interface's material. Using the receiver's material instead leaves an 83% residual — the kind of error that a structural test would never catch.

With (6) and (7), the corrected 9×9 satisfies the closed-form invariant (3) to 9.7×10^{-16} across $f = 5, 10, 25$ Hz and three interface pairs. The conventions are reconciled and the composition (1) is unblocked.

5.1 Why the existing tests could not have found this

Every test in the layered suite is structural: shape, finiteness, zero-T-no-perturbation, Born error quadratic in $\|T\|$, single-layer Foldy equals Born. *All of them pass for any self-consistent source convention*, because each is defined relative to that convention. An invariant — reciprocity, energy balance, a closed-form limit — is what discriminates. This is a recurring pattern in the project and the reason the gates recorded here are worth keeping in continuous integration rather than run on request.

6 Errata found in the thesis

Six, all now corrected in the recompiled source.

Location	Correction
Ch. 5, eq. (Rrecursion)	Reverberation operator was missing its inverse. The algorithms and the 1996 Fortran always had it; summing the round-trip series confirms the corrected form to 3×10^{-16} while the printed one is 53% wrong at 4×4.
Ch. 5, Alg. 5.1	Stale loop index $E_i \to E_1$ at the top-of-stack step.
Ch. 5, Alg. 5.3	Same slip, $E_\rho \to E_1$.
Ch. 5, UL sweeps	"Upper" and "lower" triangular labels were swapped; they also contradicted the text's own statement that the first system is solved by back substitution.
Ch. 2, eq. (Akdef)	Its shorthand $\gamma, a, b, \zeta, \chi$ is local to that equation, whereas the List of Symbols defines γ, ζ, χ as the Chapter 2 contrast ratios. Substituting those silently yields a wrong system matrix. Footnote added giving $\gamma = \lambda/(\lambda+2\mu)$, $a = 1/(\lambda+2\mu)$, $b = 1/\mu$, $\zeta = 4\mu(\lambda+\mu)/(\lambda+2\mu)$, $\chi = 2\mu\lambda/(\lambda+2\mu)$, with $\zeta - \chi = 2\mu$.

The first four are typographical. The last is the only one with teeth: the others would be caught by a careful reader, whereas the symbol collision produces a wrong matrix with nothing in the text to flag it.

7 Status and what remains

Item	Status	Evidence
Two-stage = one-stage solve	exact	3×10^{-16}
Layered Green's function	sound	translation invariance
Its reciprocity law	known	6×10^{-16}
Jump $\leftrightarrow$ force	derived	Box 5.3, validated
Corrected 9×9	reconciled	9.7×10^{-16}
Crust gradient in the reference	required	measured
Per-voxel 3-D $\mathscr{P}^x$	exists	FFT block-Toeplitz
Composed matvec	*not built*	—

What remains is implementation rather than investigation: assemble

$$\Big(\mathbf{I} - \big[\mathscr{P}^z + \mathscr{P}^x\big]\Delta\mathscr{C}_{\text{eff}}\Big)\psi$$

with $\mathscr{P}^z$ from the corrected layered propagator and $\mathscr{P}^x$ from the existing per-voxel FFT kernel, and drive it with the existing Krylov solver. The lateral operator already handles disorder — its block-Toeplitz structure comes from the regular *mesh*, not from the scatterers being identical — so no new operator is required.

All numerical values in this note are reproducible from `scripts/gate_9x9_source_convention.py` (gates A–E) and `Mathematica/BoxFiveThree.wl` (the Box 5.3 derivation).